

## **GEN 3.5 METEOROLOGICAL SERVICES**

### **3.5.1 Responsible Service**

1. The meteorological services for civil aviation are provided by the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) under the Department of Science and Technology (DOST).

The Administrator, PAGASA  
Science Garden Complex  
BIR Road, Barangay Central,  
Quezon City, Metro Manila 1100  
Philippines  
Phone: (632) 8929-4865, 3434-9040  
Email: [information@pagasa.dost.gov.ph](mailto:information@pagasa.dost.gov.ph)

or

Aviation Meteorological Services Section (AMSS)  
Weather Division, PAGASA  
Room 415, IPT BLDG, Terminal 1  
Ninoy Aquino International Airport (NAIA T1)  
Parañaque City, Metro Manila 1705  
Philippines  
Phone: (632) 8852-2927, 8879-5269, 8832-3023 (Fax)  
Email: [wd.amss@pagasa.dost.gov.ph](mailto:wd.amss@pagasa.dost.gov.ph)  
AFS: RPLLYMYX  
Website: <http://bagong.pagasa.dost.gov.ph>

2. The service is provided in accordance with the provisions contained in the following ICAO Documents:

Annex 3 - Meteorological Service for International Air Navigation

Doc. 9377 AN/915 - Manual on Coordination between Air Traffic Services, Aeronautical Information Services and Aeronautical Meteorological Services.

Doc. 9691-AN/954 - Manual on Volcanic Ash, Radioactive Material and Toxic Chemical Clouds.

3. Differences to these provisions are detailed in subsection GEN 1.7.

### **3.5.2 Area of Responsibility**

1. Area meteorological watch is provided for the Manila FIR.

## 3.5.3 Meteorological Observations and Reports

Table GEN 3.5.3 Meteorological observations and reports

| Name of station/<br>Location Indicator | Type & frequency<br>of observation/<br>automatic<br>observing<br>equipment | Types of MET<br>reports &<br>availability of<br>trend forecasts | Observation System &<br>Site(s)                                                                                                                                                              | Hours of<br>Operation | Climatological<br>information   |
|----------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------|
| 1                                      | 2                                                                          | 3                                                               | 4                                                                                                                                                                                            | 5                     | 6                               |
| BUTUAN<br>RPME                         | NIL                                                                        | METAR SPECI                                                     | Aerovane 300 M from middle<br>of RWY. Visibility estimated.<br>Cloud base estimated.                                                                                                         | H24                   | Available upon<br>prior request |
| CAGAYAN DE ORO<br>MIL<br>RPML          | NIL                                                                        | METAR SPECI                                                     | Aerovane 300 M from middle<br>of RWY. Visibility estimated.<br>Cloud base estimated.                                                                                                         | H24                   | Available upon<br>prior request |
| DIOSDADO<br>MACAPAGAL INTL<br>RPLC     | NIL                                                                        | METAR SPECI<br>Trend type<br>landing forecast                   | Aerovane, temperature,<br>RVR, ceilometer at the TDZ<br>of both ends of RWY.                                                                                                                 | H24                   | Available upon<br>prior request |
| DAET<br>RPUD                           | NIL                                                                        | METAR SPECI                                                     | Aerovane 139 M from RWY.<br>Visibility estimated. Cloud<br>base estimated.                                                                                                                   | H24                   | Available upon<br>prior request |
| DANIEL Z.<br>ROMUALDEZ<br>RPVA         | NIL                                                                        | METAR SPECI                                                     | Aerovane 170 M from RWY.<br>Visibility estimated. Cloud<br>base estimated.                                                                                                                   | H24                   | Available upon<br>prior request |
| DUMAGUETE<br>RPVD                      | NIL                                                                        | METAR SPECI                                                     | Aerovane 139 M from RWY.<br>Visibility estimated. Cloud<br>base estimated.                                                                                                                   | 2200-1000             | Available upon<br>prior request |
| FRANCISCO<br>BANGOY INTL<br>RPMO       | NIL                                                                        | METAR SPECI                                                     | Aerovane 500 M from RWY.<br>Visibility estimated. Cloud<br>base estimated.                                                                                                                   | H24                   | Available upon<br>prior request |
| ILOILO INTL<br>RPVI                    | NIL                                                                        | METAR SPECI                                                     | Aerovane 80 M from RWY.<br>Visibility estimated. Cloud<br>base estimated.                                                                                                                    | H24                   | Available upon<br>prior request |
| LAOAG INTL<br>RPLI                     | NIL                                                                        | METAR SPECI                                                     | Aerovane 650 M from THR<br>of RWY 01. Visibility<br>estimated. Cloud base<br>estimated. Thermometer<br>shelter 15 M SW of<br>anemometer post.                                                | H24                   | Available upon<br>prior request |
| LEGAZPI<br>RPLP                        | NIL                                                                        | METAR SPECI                                                     | Aerovane 150 M from RWY.<br>Visibility estimated. Cloud<br>base estimated.                                                                                                                   | H24                   | Available upon<br>prior request |
| MACTAN-CEBU INTL<br>RPVM               | NIL                                                                        | METAR SPECI<br>Trend-type<br>landing forecast                   | Aerovane, temperature,<br>RVR, ceilometer at the TDZ<br>of RWY.                                                                                                                              | H24                   | Available upon<br>prior request |
| NINYOY AQUINO INTL<br>RPLL             | NIL                                                                        | METAR SPECI<br>Trend-type<br>landing forecast                   | Aerovane are within the TDZ<br>for RWY 06/24 and RWY 13/<br>31. Ceilometer and RVR are<br>within the TDZ of RWY 06/<br>24. Temperature/dew point<br>sensors are within the TDZ<br>of RWY 06. | H24                   | Available upon<br>prior request |
| PUERTO PRINCESA<br>INTL<br>RPVP        | NIL                                                                        | METAR SPECI                                                     | Aerovane 160 M from RWY.<br>Visibility estimated. Cloud<br>base estimated.                                                                                                                   | 2200-1000             | Available upon<br>prior request |

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|----------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------|
| 1                                      | 2                                                                          | 3                                                               | 4                                                                                                                         | 5                     | 6                               |
| ROXAS<br>RPVR                          | NIL                                                                        | METAR SPECI                                                     | Aerovane 200 M from RWY.<br>Visibility and cloud base<br>estimated. Thermometer<br>shelter 106 M from center of<br>RWY.   | H24                   | Available upon<br>prior request |
| SAN JOSE<br>RPUH                       | NIL                                                                        | METAR SPECI                                                     | Aerovane 225 M from RWY.<br>Visibility and cloud base<br>estimated.                                                       | H24                   | Available upon<br>prior request |
| SUBIC BAY<br>RPLB                      | NIL                                                                        | METAR SPECI<br>Trend-type<br>landing forecast                   | Aerovane, temperature,<br>RVR, ceilometer at the TDZ<br>of both ends of RWY.                                              | H24                   | Available upon<br>prior request |
| ZAMBOANGA<br>RPMZ                      | NIL                                                                        | METAR SPECI                                                     | Aerovane 160 M from RWY.<br>Visibility and cloud base<br>estimated. Thermometer<br>shelter 15 M SW of<br>anemometer post. | H24                   | Available upon<br>prior request |

#### 3.5.4 Types of Services

Briefing and flight documentation are provided at the Aviation Meteorological Services Section (AMSS) for all flights covering the route up to the next aerodrome where briefing and flight documentation are available. In most cases, briefing is carried out using the available facilities such as the Geostationary Meteorological Satellite (GMS-5), the Integrated Runway Meteorological Observation Systems (IRMOS), the World Area Forecast System (WAFS) basic aviation products either through Internet or by a remote WAFS data transfer from a receiving station and, ICAO multi-level upper-air charts. Briefing materials include hourly METAR/SPECI not only from Manila but also from different PAGASA Aeronautical Stations.

Whenever possible the pilot-in-command or his representative is given personal briefing by a forecaster at the meteorological office, otherwise briefing may be carried out by telephone.

Meteorological aerodrome reports (METAR) and forecasts, including trend-type forecasts, are provided on a routine basis at the AMSS.

En-route information is normally presented in the form of a prognostic charts:

- Significant Weather Chart (FL250-FL630) with 500 HPA-level ascend/descend chart inscribed;
- 300 HPA (FL300) chart;
- 200 HPA (FL300) chart; and
- Forecast take-off conditions at Manila, i.e. temperature and altimeter setting, altogether in a compilation page of Terminal Aerodrome Forecast (TAFOR) for the intended airports of destination, alternate aerodrome, take-off alternate, en-route alternate and destination alternate.

#### 3.5.5 Notification Required from Operators

Notification from operators with respect to flight documentation and other meteorological information needed by them is normally required at least 12 hours before aircraft estimated time of departure. Briefing is available anytime.

### 3.5.6 Aircraft Operators

- Pursuant to ICAO Annex 3, Meteorological Service for International Air Navigation, the making and transmission of aircraft observations (AIREP) are required at the following compulsory MET reporting points within the Manila FIR.

| ATS ROUTE               | ATS/MET REPORTING POINT | COORDINATES      |
|-------------------------|-------------------------|------------------|
| A341                    | VIMAG                   | 060136N 1183236E |
| A461                    | NITIP                   | 040000N 1234306E |
| A461 / M501             | NOMAN                   | 200000N 1164018E |
| A582                    | BISIG                   | 210000N 1251600E |
| A583 / L625 / M754 / W3 | AKOTA                   | 162743N 1171224E |
| A583                    | SABNO                   | 185906N 1155042E |
| A590 / G578             | GURAG                   | 210000N 1272500E |
| A590 / M501 / R597      | VIGOR                   | 172211N 1242537E |
| M646                    | OSANU                   | 074124N 1171736E |
| M646                    | POTIB                   | 210000N 1204530E |
| B462                    | BIDOR                   | 040000N 1313000E |
| B462 / L625             | MEVIN                   | 210000N 1223300E |
| B473                    | ELNIT                   | 040000N 1280500E |
| G467 / P515             | ENDAX                   | 141500N 1300000E |
| G578                    | GORAI                   | 040000N 1204400E |
| L625 / L628             | ARESI                   | 135824N 1142700E |
| M501                    | LIMLE                   | 163942N 1300000E |
| M754                    | VINIK                   | 083830N 1161348E |
| M765 / W17              | PANDI                   | 113806N 1140000E |
| M772 / N884             | LAXOR                   | 094937N 1144829E |
| N884                    | LEBIX                   | 210000N 1235430E |
| N892                    | KABAM                   | 210000N 1195248E |
| R337                    | ISGOG                   | 091442N 1300000E |

The AIREP made during the preceding stage of all flights arriving at the Ninoy Aquino International Airport (NAIA) shall be delivered by the operator without delay to the Aviation Meteorological Services Section, Room 415 IPT BLDG, Terminal 1, NAIA.

### 3.5.7

### Table GEN 3.5.7 VOLMET service

| Name of Station | CALL SIGN/<br>IDENT/<br>Abbreviation<br>(EM) | Frequency | Broadcast<br>period | Hours of service | Aerodromes/<br>Heliports<br>included | REP, SIGMET<br>INFO, FCST &<br>Remarks |
|-----------------|----------------------------------------------|-----------|---------------------|------------------|--------------------------------------|----------------------------------------|
| 1               | 2                                            | 3         | 4                   | 5                | 6                                    | 7                                      |
| Nil             | Nil                                          | Nil       | Nil                 | Nil              | Nil                                  | Nil                                    |

3.5.8 SIGMET and AIRMET Service

Table GEN 3.5.8 SIGMET and AIRMET service

| Name of MWO/<br>location indicators | Hours | FIR or CTA served | Validity | Specific SIGMET procedures | AIRMET Procedures | ATS unit served | Additional information |
|-------------------------------------|-------|-------------------|----------|----------------------------|-------------------|-----------------|------------------------|
| 1                                   | 2     | 3                 | 4        | 5                          | 6                 | 7               | 8                      |
| Nil                                 | Nil   | Nil               | Nil      | Nil                        | Nil               | Nil             | Nil                    |

**1. General**

For the safety of air traffic, the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) maintains an area meteorological watch and warning service. This service consists partly of a continuous weather watch within the FIR and the issuance of appropriate information (SIGMET) by Meteorological Watch Offices (MWO) and partly, issuing of warnings for the respective aerodrome and, subject to agreement, for other aerodromes by all aeronautical MET offices.

**2. Area meteorological watch service**

The area meteorological watch service is performed by the Aviation Meteorological Services Section (AMSS). The PAGASA also maintains a number of aerodrome meteorological watch offices strategically established within the FIR.

The AMSS, as MWO, issue information in the form of SIGMET messages about the occurrence or expected occurrence of one or both of the following significant meteorological phenomena:

- tropical cyclone
- volcanic ash cloud
- and other severe weather advisories like monsoons

The SIGMETs are issued in abbreviations and plain language using ICAO abbreviations and are chronologically numbered. Final issuance is done whenever the disturbance makes an exit with the FIR or dissipation occurs within the FIR. The period of validity is generally limited to not more than six hours with extended outlook of 24 hours in the case of tropical cyclones and less than four hours with outlook of 12 hours for volcanic ash cloud. SIGMET for volcanic ash cloud is issued as frequent as needed.

The AMSS transmit SIGMETs to local MET offices, to the regional control centers competent for the FIR or UIR concerned, in like manner to the following users: operators, flight crew members, air traffic services units, search and rescue services units, airport management and upon agreement, also SIGMETs from adjoining Volcanic Ash Advisory Center (VAAC).

**3. Warning service**

Warning in the form of advisory or bulletin, for the protection of parked or moored aircraft or of other equipment at the airport are issued by all meteorological offices, if occurrence of tropical cyclone or other inclement weather is expected to affect respective aerodromes.

The warnings are generally issued in English and are distributed in accordance with a distribution list which had been agreed upon locally. In order to guarantee efficient and prompt dissemination of the warnings, the distribution list to be used shall, as far as possible, contain only one recipient for an interested group, this recipient will be responsible for further dissemination of the warning within the group.

SIGMET information is disseminated, in addition to directed transmissions to aircraft general calls, as an aeronautical radio broadcast between 0700Z until SS + 30.

- a. by the Manila Aeronautical Mobile Station (Manila Radio) for Manila FIR.
- b. by the ATS units for their own area of responsibility.

The information is broadcast from the MWO concerned and repeated every half and full hour during the period of validity of the SIGMET information.

3.5.9 Other Automated Meteorological Services

Table GEN 3.5.9 Other automated meteorological services

| Service name | Information available | Area, route and aerodrome coverage | Telephone, telefax numbers, Email and Website Remarks |
|--------------|-----------------------|------------------------------------|-------------------------------------------------------|
| 1            | 2                     | 3                                  | 4                                                     |
| Nil          | Nil                   | Nil                                | Nil                                                   |